Supplementary Material: Within- and between-year seasonal patterns of respiratory pathogens and the potential for concurrent outbreaks

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Abstract

Supplementary materials for the main manuscript.

Keywords: respiratory pathogens, influenza, RSV, seasonal dynamics, survival analysis, concurrent outbreaks, geographic variation

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1. Supplemental Materials

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1.1. Tables

1.1.1. Comparable Models by City and Pathogen

Table 1: Table S1: Number of comparable models (ΔAIC < 7) per city and pathogen. This table illustrates the distribution of model availability across the eight study cities, highlighting the variation in the number of comparable models per city.

City	**Influenza**	**RSV**
North		
Beijing	20	39
Lanzhou	2	4
Xian	2	3
North Total	*24*	*46*
South		
Wuhan	20	22
Suzhou	18	3
Guangzhou	4	11
Wenzhou	2	2
Yunfu	2	2
South Total	*46*	*40*
Grand Total	**70**	**86**

1.1.2. Median Probability of Concurrent Outbreaks by Risk Threshold

Table 2: Table S2: Median probability of concurrent outbreaks by city and risk threshold. Values represent the median proportion of years with concurrent high-risk periods for both Influenza and RSV across 1000 simulation paths. Risk thresholds are defined as joint top 30%, 20%, and 10% percentiles of disease burden.

City	**Top 30%**	**Top 20%**	**Top 10%**
North			
Beijing	0.200	0.100	0.000
Lanzhou	0.118	0.059	0.000
Xian	0.450	0.350	0.150
South			
Wuhan	0.400	0.300	0.150
Suzhou	0.350	0.200	0.100
Guangzhou	0.188	0.125	0.062
Wenzhou	0.812	0.500	0.250
Yunfu	0.375	0.188	0.125

1.1.3. Median Time to Next Concurrent Outbreak by Risk Threshold

Table 3: Table S3: Median time to subsequent concurrent outbreak (years) by city and risk threshold, with 95% confidence intervals. Values represent the x-intercepts of the dashed vertical lines in Figure 6, indicating the time (in years) at which 50% of simulation paths are expected to experience their subsequent concurrent outbreak. Confidence intervals (in parentheses) reflect uncertainty in the Kaplan-Meier survival estimates, which properly account for right-censoring of paths that never experience a second outbreak. The relatively narrow confidence intervals reflect high precision from aggregating across 1000 simulation paths per city. Values marked as "—" indicate that the 50% threshold was not reached within the simulation window. Risk thresholds are defined as joint top 30%, 20%, and 10% percentiles of disease burden.

City	**Top 30%**	**Top 20%**	**Top 10%**
North			
Beijing	3.0 (2.0–3.0)	5.0 (4.0–6.0)	—
Lanzhou	5.0 (5.0–6.0)	—	—
Xian	1.0 (1.0–1.0)	2.0 (2.0–2.0)	3.0 (3.0–3.0)
South			
Wuhan	1.0 (1.0–1.0)	2.0 (2.0–2.0)	3.0 (3.0–3.0)
Suzhou	2.0 (2.0–2.0)	3.0 (3.0–3.0)	6.0 (5.0–6.0)
Guangzhou	3.0 (3.0–4.0)	6.0 (6.0–7.0)	—
Wenzhou	1.0 (1.0–1.0)	1.0 (1.0–1.0)	2.0 (2.0–2.0)
Yunfu	2.0 (2.0–2.0)	3.0 (3.0–3.0)	7.0 (6.0–8.0)

1.1.4. Optimal Within-Year and Between-Year Cycle Lengths

Table 4: Table S4: Optimal within-year and between-year cycle lengths (months) for each city and pathogen combination. Values represent the cycle lengths of the best-performing model (lowest AIC) for each city-pathogen pair. Within-year cycles are shown for both pathogens; between-year cycles are only shown for models that included multi-seasonal decomposition (MSTL). These values correspond to the data presented in Figure 3 (A) and (B) of the main text.

Region	**City**	**Within-year cycle (months)**		**Between-year cycle (months)**	
		Influenza	**RSV**	**Influenza**	**RSV**
North					
	Beijing	13.0	10.0	28.0	24.0
	Lanzhou	13.0	13.0	20.0	35.0
	Xian	13.0	13.0	36.0	33.0
South					
	Suzhou	14.0	12.0	26.0	21.0
	Wuhan	13.0	12.0	30.0	19.0
	Wenzhou	12.0	12.0	22.0	18.0
	Guangzhou	13.0	12.0	34.0	31.0
	Yunfu	14.0	10.0	21.0	24.0

1.2. Figures

1.2.1. Distribution of Between-Year Cycle Lengths

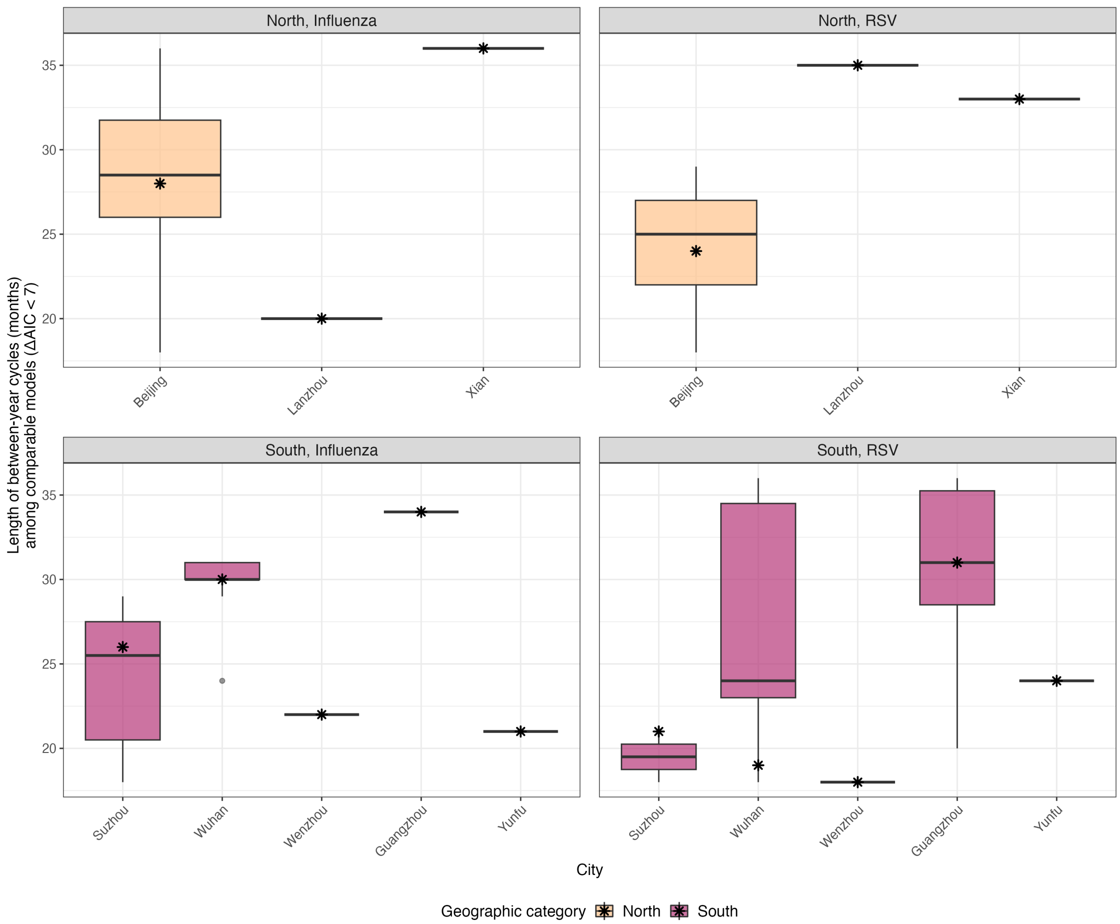

Figure 1: Figure S1: Distribution of between-year cycle lengths (months) among comparable models (ΔAIC < 7) by city and pathogen, separated by geographic region (North vs South). Each panel displays boxplots of the between-year cycle lengths (in months) for each city within the specified region and pathogen combination. The asterisk (*) marks the best-performing model (lowest AIC) for each city-pathogen combination.

Supplementary Text 1. Literature search strategy

We identified city-level surveillance data for respiratory syncytial virus (RSV) and influenza virus (InfV) through structured searches in PubMed. Searches were conducted on 12 August 2024, with no language restrictions. The full search strings and screening criteria are provided below.

RSV search

The PubMed search string was:

```
("Respiratory Syncytial Viruses"[Mesh] OR RSV[Title/Abstract]
 OR "respiratory syncytial"[Title/Abstract])
AND (China[Mesh] OR China[Title/Abstract] OR Chinese[Title/Abstract])
```

Studies were included if they met all of the following criteria:

- Surveillance data reported at the prefecture- or city-level (administrative level two);
- At least five consecutive years of observations;
- Monthly or weekly temporal resolution;
- Laboratory-confirmed RSV indicators (positivity among ARI/LRTI inpatients or outpatients).

Through this process, eight cities were identified with eligible RSV time series: Beijing, Lanzhou, Xi'an, Wuhan, Suzhou, Wenzhou, Guangzhou, and Yunfu.

Influenza search

For the same eight cities, we applied a parallel PubMed search using the following query:

```
("Influenza, Human"[Mesh] OR influenza[Title/Abstract])
AND (China[Mesh] OR China[Title/Abstract] OR Chinese[Title/Abstract])
AND "last 10 years"[PDat]
```

The same inclusion criteria were applied (prefecture-level data, $\geq$5 years of continuous observations, and monthly or weekly reporting). For cities with multiple eligible influenza series, we selected the dataset with the longest uninterrupted coverage and the most specific virological definition. Data sources corresponding to each city–pathogen pair are listed in Supplementary Table S1.

Table 5: Supplementary Table S1. Surveillance indicator definitions for RSV and influenza (InfV)

City	Pathogen	Indicator definition	Clinical setting	Labor
Beijing	RSV	RSV antigen positivity in ARI inpatients	Inpatient (ARI)	IFA/I
Beijing	InfV	Influenza-positive samples by subtype	Outpatient (ILI sentinel)	Virus
Xi'an	RSV	RSV-positive cases / positivity rate	Out-/inpatient (ARI)	Not r
Xi'an	InfV	Daily ILI case counts (syndromic)	Outpatient (ILI)	None
Lanzhou	RSV	RSV positivity; RSV-A genotypes	Inpatient (ARI)	RT-P
Lanzhou	InfV	Laboratory-confirmed influenza	Out-/inpatient	RT-P
Wuhan	RSV	RSV positivity in ARTI inpatients	Inpatient (ARTI)	DFA
Wuhan	InfV	ILI% and influenza-positive specimens	Outpatient (ILI sentinel)	RT-P
Wenzhou	RSV	RSV-positive LRTI inpatients	Inpatient (LRTI)	DFA
Wenzhou	InfV	Influenza-positive LRTI inpatients	Inpatient (LRTI)	DFA
Suzhou	RSV	RSV positivity in neonatal LRTI inpatients	Inpatient (neonatal LRTI)	DFA
Suzhou	InfV	Influenza positivity in SARI inpatients <5y	Inpatient (SARI)	RT-P
Guangzhou	RSV	RSV positivity in ARI patients	In-/outpatient (ARI)	qRT-I
Guangzhou	InfV	ILI%, influenza positivity, ILI+ proxy	Outpatient (ILI)	RT-P
Yunfu	RSV	RSV positivity in hospitalized children	Inpatient (pediatric ARI)	DFA
Yunfu	InfV	Influenza positivity in ILI outpatients	Outpatient (ILI sentinel)	Isolat

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